

Ashton E. Thomas

EDUCATION

University of Michigan - Rackham Graduate School <i>Major - MSE Computer Engineering — Concentration in ML/CV</i>	08/2025 – 05/2026 Ann Arbor, MI, USA
University of Michigan College of Engineering <i>Major - BSE Computer Science, GPA - 3.8 / 4.0</i> <i>Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics</i> <i>Data Structures & Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems</i>	08/2022 – 05/2025 Ann Arbor, MI, USA

EXPERIENCE

Amazon <i>Software Development Engineer I, AWS</i> - Build developer-insights platforms at AWS for software delivery health, observability, and engineering analytics as part of a 12+ person team. - Completed an SDE internship on Alexa endpoint experiences, optimizing customer-facing device functionality with a 20+ engineer team using React Native, Kotlin, and TypeScript. - Managed Amazon company-wide employee data for 100,000s of employees as part of a 10-engineer team, including collection, processing, and executive-level inferences.	09/2024 – Present Boston, MA & New York, NY, USA
Ground Vehicle Systems Center (SEC) <i>Software Development Intern</i> - Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency. - Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.	05/2024 – 07/2024 Warren, MI, USA
Madi Taylor Photo <i>Full Stack Developer</i> - Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS. - Implemented robust back-end payment solutions and form validation to streamline user transactions.	06/2022 – 07/2024 Traverse City, MI, USA

RESEARCH

Polk Lab — University of Michigan <i>Computational Neuroscience Research Assistant</i> - Collaborated with a PhD student to develop a machine learning model aimed at mimicking neural distinctiveness of the human brain, while also enabling the separation of background noise from spoken language. - Worked alongside Prof. Thad Polk to develop a script that processes CSV files with 1000s of columns, utilizing wildcard parsing to dynamically filter columns. The script includes a node hierarchy for efficient column interrelation and supports operations such as adding, removing, printing, and row filtration.	09/2024 – 06/2025 Ann Arbor, MI, USA
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PROJECTS

Geoguessr AI, Computer Vision - Designed and implemented a modified ResNet-50 architecture for geographic location identification from images. - Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ~90% accuracy.	2023 – 2024
Virtual Memory Pager, Operating Systems - Built a virtual-memory pager managing per-process arenas, page tables, physical frames, swap-backed pages, and shared file-backed mappings. - Implemented page-fault handling, copy-on-write behavior, dirty/reference tracking, swap allocation, file-block sharing, eviction, and a clock-style page replacement policy.	2025
Invariant Learning Experiments, Machine Learning - Compared empirical risk minimization, invariant risk minimization, and IRM-Games under distribution shift across controlled experimental environments. - Evaluated model behavior on Colored MNIST, synthetic chain-equation data, and financial time-series settings with spurious correlations, trend regimes, and volatility shifts.	2025

TECHNICAL SKILLS

ARM, ARCGIS, AWS, Lambda, S3, Glue, Athena, CloudWatch, CDK, C#, C++, CSS, Excel, Flask, Git, HTML, Javascript, Jinja, Jira, Keras, Kotlin, Latex, MagicDraw, Matlab, NumPy, Pandas, Python, PyTorch, Qiskit, React/Native, SQL, Typescript, VSCode